Exchange Monte Carlo simulation of an Ising spin glass

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Abstract

This project implements the Exchange Monte Carlo (Parallel Tempering) algorithm in Fortran for the Ising spin glass with Hamiltonian $H(S) = -\sum_{\langle i,j \rangle} J_{ij} S_i S_j$, where J_{ij} are independent Gaussian random variables with zero mean and unit variance, and $S_i = \pm 1$. The code is first validated with the exact Ferdinand–Fisher results for the 2D ferromagnetic Ising model ($J_{ij} = 1$, $d = 2$, $L = 20$). Production runs are then executed for the 3D Gaussian spin glass ($L = 4$ and $L = 8$), computing the averaged disorder overlap distribution $P(q)$, $\langle q^2 \rangle$, and $\langle E \rangle / N$ as functions of temperature. Additionally, a comparison with a single Metropolis simulation at $T = 0.2$ was done, as well as an autocorrelation time analysis, confirm the efficiency advantage of parallel tempering at low temperatures. [1]

1 Estimates, interpretations and comments

1.1 Initial test and validation against Ferdinand–Fisher ($d = 2$, $L = 20$)

First, a simulation with $J_{ij} = 1$ (ferromagnet), $d = 2$, $L = 20$, $N_T = 21$ temperatures linearly spaced in [1.0, 3.0], and $n_{\mathrm{MCS}} = 5 \times 10^6$ was run. The statistical errors were later extracted using a binning analysis, using the plateau in the error vs. block-size curve (Fig. 2) to determine the block size at which autocorrelations are resolved.

For instance, at $T = 2.0$ the estimate is $\langle E\rangle/L^2 = -1.74571 \pm 0.0002$, in agreement with the Ferdinand–Fisher exact value $\langle E\rangle/L^2\big|_{\mathrm{FF}} = -1.74555\ldots$ to within the statistical error. Agreement is equally good across the remaining temperatures in this range (Fig. 1), confirming the correctness of the implementation.

All exchange acceptance rates remain comfortably above the required minimum of 0.3: the lowest pair ($T = 2.9$–3.0) has a rate of ≈ 0.32, while the highest ($T = 1.0$–1.1) reaches ≈ 0.87 (Fig. 3). The overlap time series at representative temperatures (Fig. 4) show rapid decorrelation at high T and slow (and with large amplitude) fluctuations near the critical temperature, as expected.

1.2 Overlap distribution $P(q)$ for the 3D spin glass ($d = 3$, $L = 4$, 1000 samples)

The overlap $q = Q/N$ is measured for 1000 independent disorder realizations, each simulated for at least 10^4 MCS, discarding the first 3×10^3 MCS as thermalization. Since the Hamiltonian is invariant under a global spin flip, $P(q)$ is symmetric in q, so following Katzgraber et al. [1], the distribution is plotted in $q \in [0, 1]$.

At both temperatures, $P(q)$ shows a pronounced peak at large q and a tail extending down to $q = 0$, in qualitative agreement (the overal shape) with Figs. 2 and 3 of Katzgraber et al. [1] (Fig. 5 here). At $T = 0.5$ the peak is broad, and the weight at small q is significant, showing the presence of many metastable states with a wide range of overlaps. At $T = 0.2$, further in the spin-glass phase, the peak gets sharper and shifts toward $q \approx 1$ and the tail at small q persists. This is inconsistent with the droplet theory that predicts a finite and approximately size independent $P(0)$, and consistent with the replica simmetry breaking case.

1.3 Convergence to equilibrium ($d = 3$, $L = 8$, 5 samples)

Five independent disorder samples were simulated for 10^6 MCS with $N_T = 32$ temperatures in [0.2, 2.0]. The binning errors of $\langle q^2\rangle$ and $\langle E\rangle/N$ were computed in four successive time windows: $[10^4 : 3 \times 10^4]$, $[3 \times 10^4 : 10^5]$, $[10^5 : 3 \times 10^5]$, and $[3 \times 10^5 : 10^6]$ MCS (Figs. 6 and 7).

At high temperatures ($T \gtrsim 1.0$) the errors are small and stable across all windows. At low temperatures the errors decrease from the first to the last window, indicating that the system is still approaching equilibrium and by the final window the errors have converged for $T \gtrsim 0.3$. At $T = 0.2$ the errors are still larger in the first window than in the last, and do not stabilize across windows, indicating that 10^6 MCS might not be enough to equilibrate the system at this temperature.

This is the expected behavior in the spin-glass phase: at low temperatures, the free-energy barriers between metastable states grow, making equilibration increasingly slow [1].

1.4 Equilibrium observables, last time interval ($d = 3$, $L = 8$)

Using only the data from the last window $[3 \times 10^5 : 10^6]$ MCS, $\langle q^2\rangle$ and $\langle E\rangle/N$ are plotted as functions of temperature (Figs. 8 and 9) and $P(q)$ for $T = 0.2, 0.5, 1.0, 2.0$ (Figs. 10–13).

At high temperatures $\langle q^2\rangle \to 0$, consistent with independent, uncorrelated replicas in the paramagnetic phase. Below $T \approx 1.0$ the overlap grows rapidly, reaching $\langle q^2\rangle = 0.811 \pm 0.085$ at $T = 0.2$ (Table 1). There is a large dispersion of samples, that is consistent with the expected behavior of a spin glass: in the spin glass phase, the overlap is strongly dependent on disorder, so with only 5 samples the mean is not precisely determined. The energy $\langle E\rangle/N$ decreases monotonically with temperature, as expected.

The $P(q)$ plots confirm the expected behaviour: a narrow Gaussian at $T = 2.0$ evolves into a wide and flat distribution at $T = 0.5$ and a bimodal distribution at $T = 0.2$, consistent with a low temperature phase with replica symmetry breaking. The shape of $P(q)$ changes significantly from sample to sample at low T, while at $T = 2.0$ all samples give essentially the same Gaussian.

Table 1: Disorder-averaged $\langle q^2 \rangle$ and $\langle E \rangle / N$ from the last time window ($L = 8$, 5 samples). σ_s: standard deviation over samples; $\sigma_{\bar{x}}$: standard error of the mean.

T	$\langle q^2 \rangle$	$\sigma_s(q^2)$	$\langle E \rangle / N$	$\sigma_s(E/N)$
0.200	0.811	0.190	−1.683	0.016
0.374	0.601	0.197	−1.677	0.016
0.548	0.520	0.200	−1.664	0.017
0.722	0.413	0.200	−1.642	0.019
1.013	0.179	0.145	−1.584	0.025
1.500	0.057	0.024	−1.462	0.019
2.000	0.022	0.005	−1.287	0.010

1.5 Comparison with Metropolis at $T = 0.2$ (optional)

A single-temperature Metropolis simulation was run at $T = 0.2$ for the same 5 samples, also for 10^6 MCS. Figure 14 shows the running average of $\langle q^2 \rangle$ as a function of MC time for both algorithms. Parallel tempering converges to its equilibrium value ≈ 0.81 in the first few 10^3 MCS, but the Metropolis average remains close to zero throughout the entire 10^6 MCS run, failing to equilibrate. This is the expected consequence of exponentially large free-energy barriers at low temperatures: Metropolis cannot escape from its initial metastable configuration, while parallel tempering allows replicas to travel to high temperatures where barriers vanish. [1]

1.6 Autocorrelation times (optional)

The normalised autocorrelation function $\rho(t)$ and integrated autocorrelation time

$$\tau_{\text{int}} = \frac{1}{2} + \sum_{t=1}^{t_{\max}} \rho(t) \tag{1}$$

were computed for both q and E/N at $T = 0.2$ (Figs. 15, 16, 17). Results are given in Table 2.

Table 2: Integrated autocorrelation times τ_{int} (measurement steps) at $T = 0.2$, averaged over 5 samples ($\pm$ std).

Method	$\tau_{\text{int}}(q)$	$\sigma[\tau(q)]$	$\tau_{\text{int}}(E)$	$\sigma[\tau(E)]$
Parallel Tempering	19.9	8.8	48.3	10.2
Metropolis	141.6	18.0	166.8	4.8
Ratio (Met/PT)	7.1		3.5	

Parallel tempering reduces the autocorrelation time of q by a factor of ≈ 7 and that of E by a factor of ≈ 3.5 compared to Metropolis at $T = 0.2$. But these ratios underestimate the true advantage: the Metropolis run has not equilibrated (Section 1.5), so its effective autocorrelation time is, in practice, infinite for this system size.

1.7 Comments

The most instructive aspect of the project was observing directly how Metropolis fails to equilibrate at low temperature while parallel tempering does so rapidly, which can be seen both in the convergence of running averages (Part 6) and in the autocorrelation times (Part 7).

A key implementation decision was swapping the temperature labels assigned to replicas rather than copying all spin configurations at every exchange attempt. In the code, this means tracking two index arrays, `rep_at_temp(k)` and `temp_of_rep(r)`, but avoids an $\mathcal{O}(N)$ memory copy. Also, a non-trivial bug was found in `bonds.f90`: the subroutine `set_gaussian_bonds()` was overwriting J_{ij} twice (once when looping from site i and again when the same pair was visited from site j) effectively reassigning a new Gaussian value and breaking the symmetry $J_{ij} = J_{ji}$.

The binning subroutine was adapted from the one developed in the MoMo course. Extending it to Part 4, with multiple time windows and multiple samples, required a dedicated `main_binning_part4.f90`. Also, one of the persistent difficulties was choosing N_T large enough that all swap acceptance rates exceeded 0.3 across the full range [0.2, 2.0]. In practice, that meant choosing a much denser temperature grid at low T than at high T.

Time series files for Part 3 quickly exceeded both the 30 MB guideline and GitHub's file size limit. The solution was to compute overlap histograms directly at runtime in `main3.f90` rather than writing full time series to disk, which also motivated the split between `main.f90` (Parts 1, 2, 4, 5) and `main3.f90` (Part 3). For Parts 4 and 6 the time-series files were compressed before committing to the repository. In retrospect, simply reducing the output frequency would have been a much simpler and also valid solution.

Additionally, computing autocorrelation functions for all samples and both algorithms required looping over ten input files. The execution loop was integrated into the Makefile as the `autocorr7` target, following a suggestion from GitHub Copilot, to avoid running the autocorrelation executable manually for each file. More broadly, Copilot was used as a debugging and suggestion tool in both Fortran and Python. I declare that all physical reasoning, algorithmic choices, and interpretation of results are the author's own (myself, colleagues and cited references). AI assistance is flagged explicitly in the relevant source files.

2 Figures

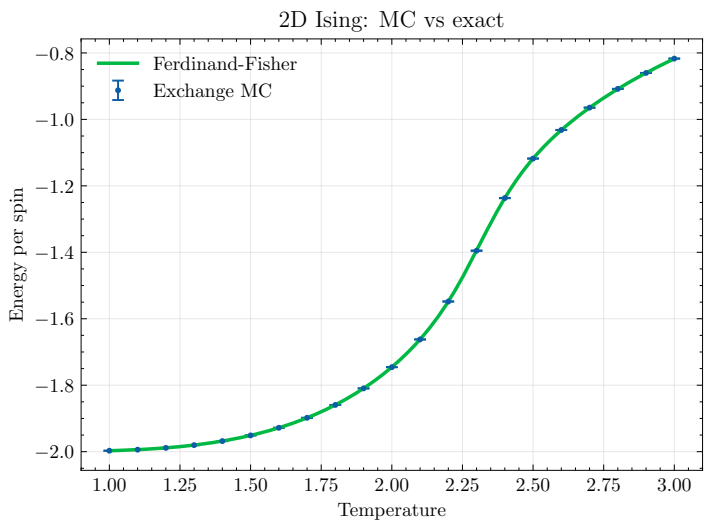

Figure 1: Mean energy per spin $\langle E \rangle / N$ vs. temperature for the 2D ferromagnetic Ising model ($L = 20$, $N_T = 21$), compared with the exact Ferdinand–Fisher result. Error bars from binning analysis.

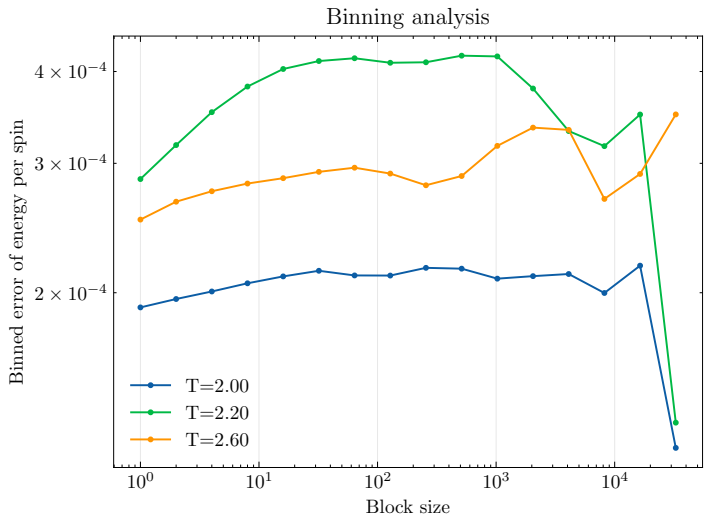

Figure 2: Binning error of $\langle E \rangle / N$ as a function of block size at representative temperatures ($d = 2$, $L = 20$). The plateau gives the corrected statistical error.

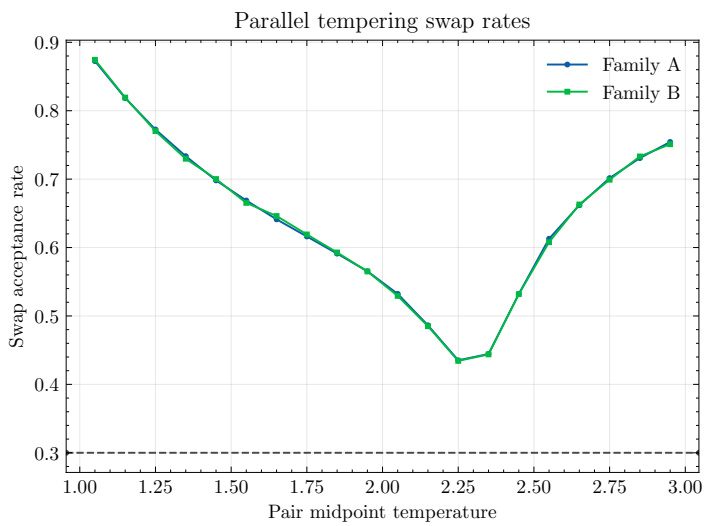

Figure 3: Exchange acceptance rates for neighbouring temperature pairs ($T \in [1.0, 3.0]$, $N_T = 21$). Both replica families a and b are shown.

References

References

[1] H. G. Katzgraber, M. Palassini, and A. P. Young, "Monte Carlo simulations of spin glasses at low temperatures", *Phys. Rev. B* **63**, 184422 (2001).

[2] M. Palassini, "Exchange Monte Carlo (Parallel Tempering)", Advanced Simulation Methods (MASM), Spring 2026 (lecture notes provided on Campus Virtual).

Code: GitHub repository

`https://github.com/itxasoma/masm-exchangeMC`

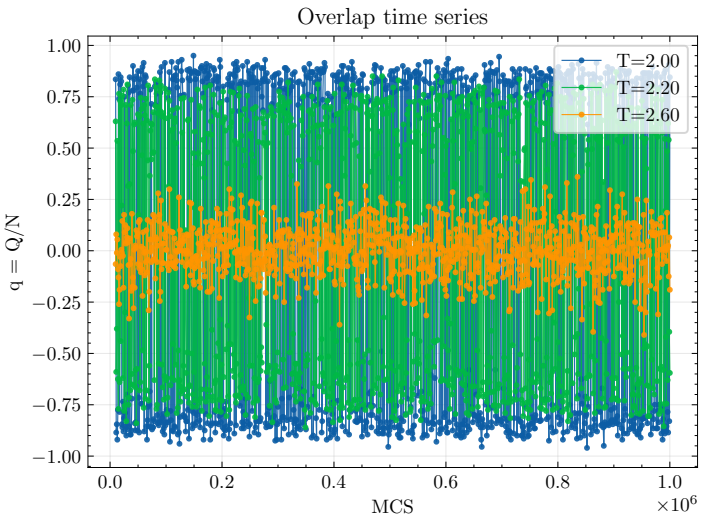

Figure 4: Time series of the normalised overlap $q = Q/N$ at representative temperatures ($d = 2$, $L = 20$).

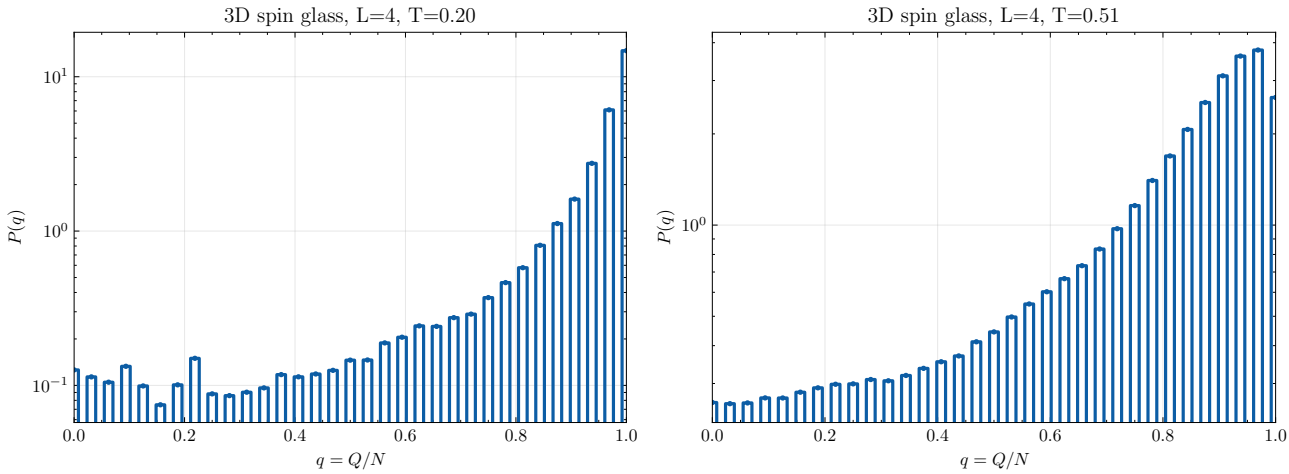

Figure 5: Disorder-averaged overlap distributions $P(q)$ at $T = 0.2$ and $T = 0.5$ ($d = 3$, $L = 4$, 1000 samples).

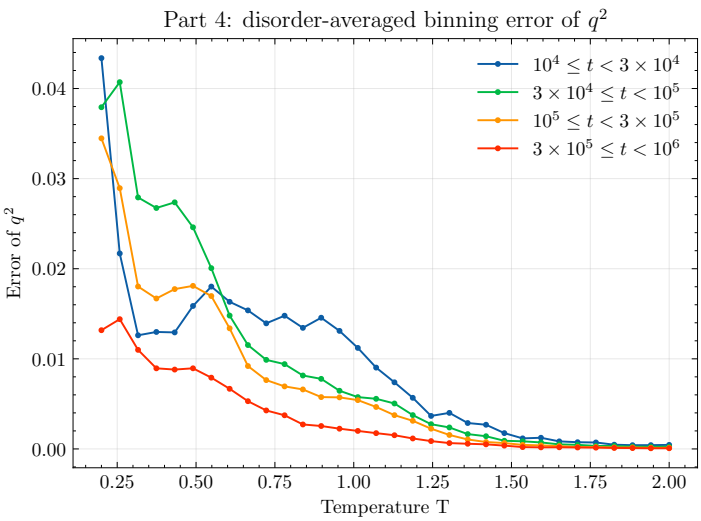

Figure 6: Binning error of $\langle q^2 \rangle$ vs. temperature for the four successive time windows ($d = 3$, $L = 8$, 5 samples). Decreasing errors with later windows indicate convergence toward equilibrium.

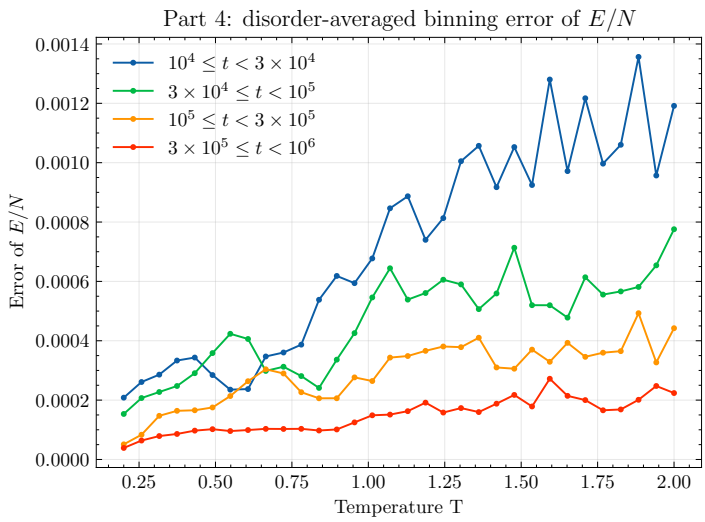

Figure 7: Binning error of $\langle E\rangle/N$ vs. temperature for the four time windows ($d = 3$, $L = 8$, 5 samples).

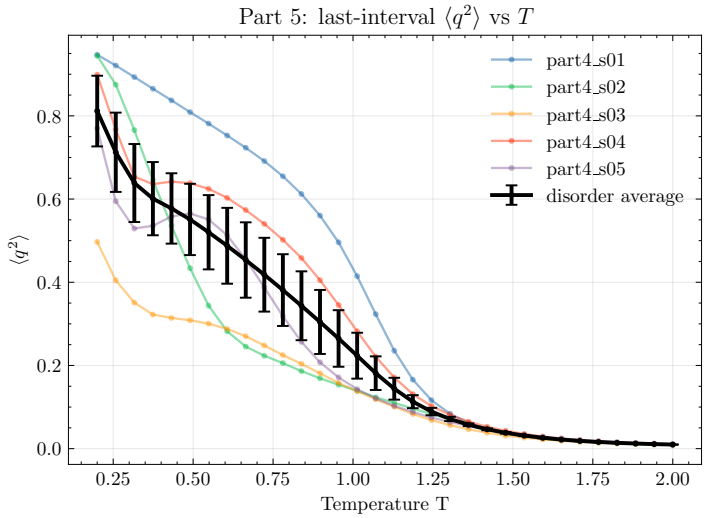

Figure 8: Disorder-averaged $\langle q^2\rangle$ vs. temperature, last time window $[3\times10^5 : 10^6]$ MCS ($L = 8$, 5 samples). Thin lines: individual samples; thick black line: disorder average with error bars.

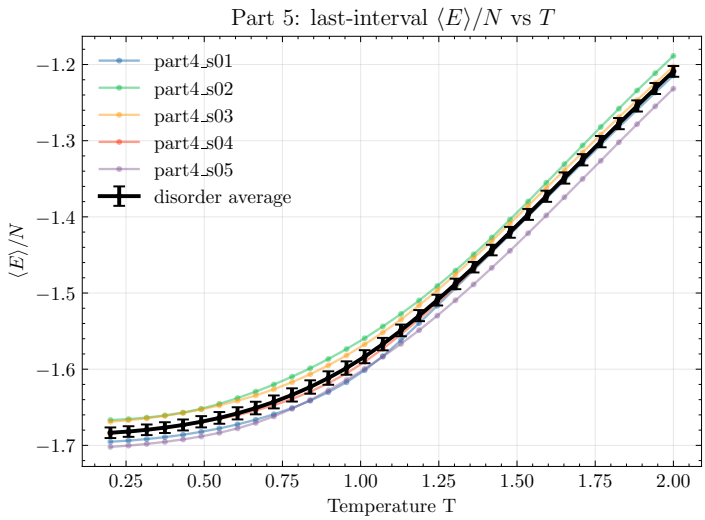

Figure 9: Disorder-averaged $\langle E\rangle/N$ vs. temperature, last time window ($L = 8$, 5 samples).

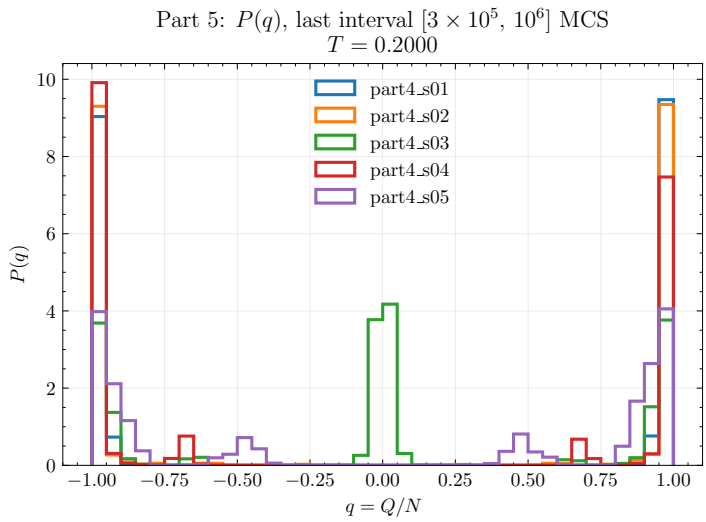

Figure 10: $P(q)$ at $T = 0.2$ ($L = 8$, 5 samples, last time window). Individual samples shown as thin lines; average as thick line.

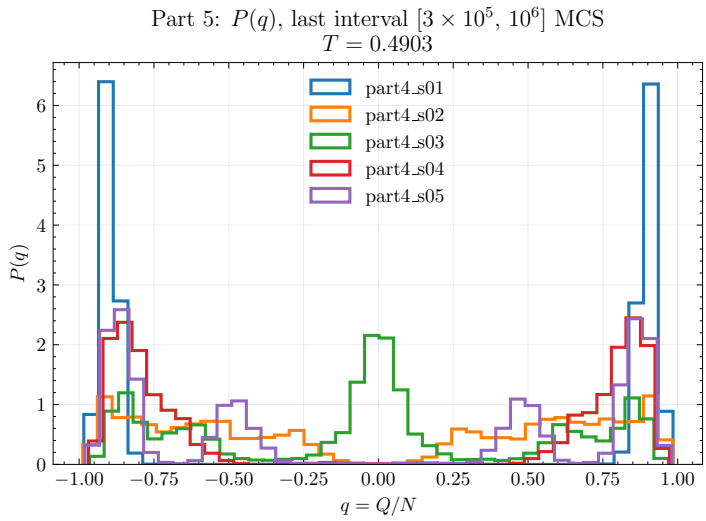

Figure 11: $P(q)$ at $T = 0.5$ ($L = 8$, 5 samples).

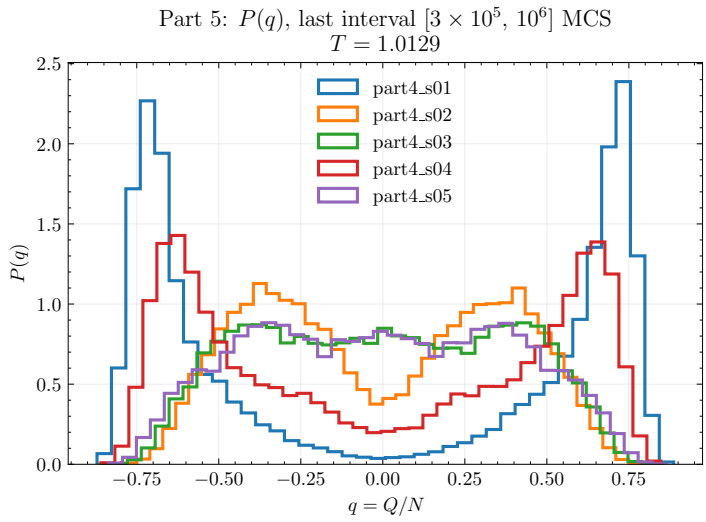

Figure 12: $P(q)$ at $T = 1.0$ ($L = 8$, 5 samples).

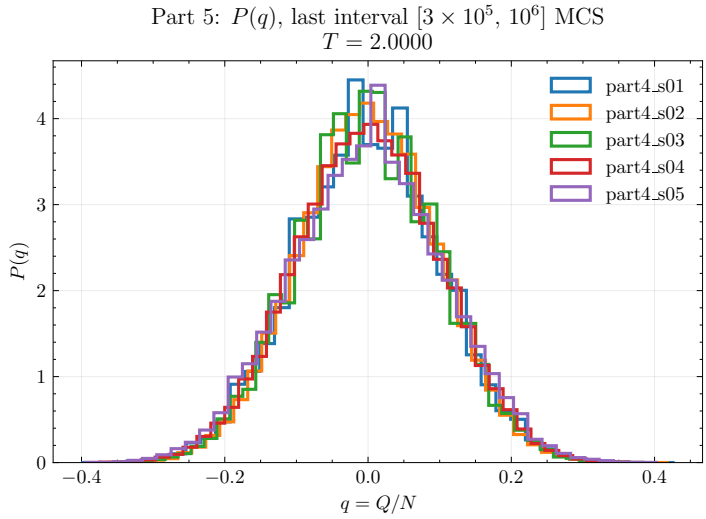

Figure 13: $P(q)$ at $T = 2.0$ ($L = 8$, 5 samples). The narrow Gaussian at $q = 0$ is consistent with the paramagnetic phase.

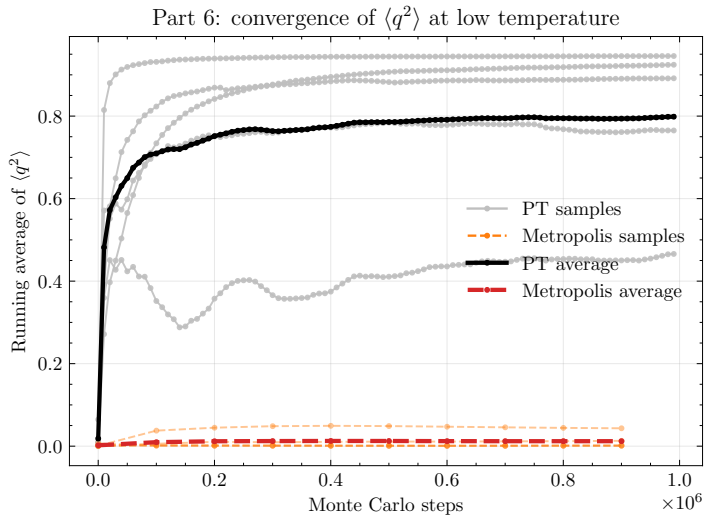

Figure 14: Running average of $\langle q^2 \rangle$ vs. MC time at $T = 0.2$: Parallel Tempering (Part 4) vs. Metropolis (Part 6), 5 disorder samples. PT converges rapidly; Metropolis fails to equilibrate within 10^6 MCS.

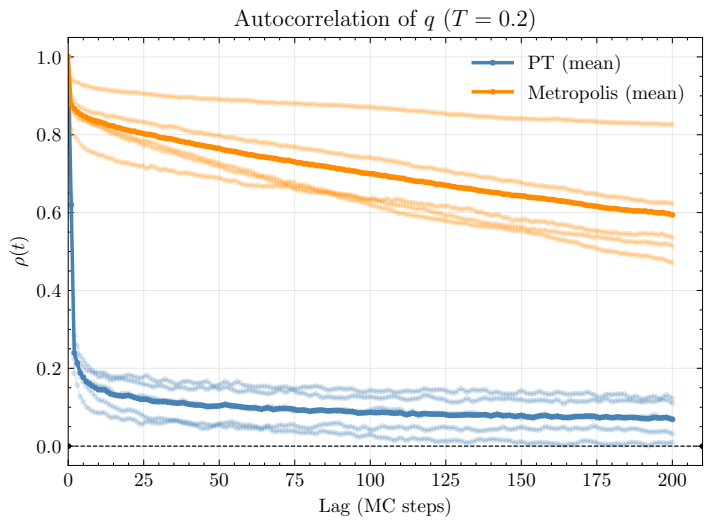

Figure 15: Normalised autocorrelation function $\rho(t)$ of the overlap q at $T = 0.2$: PT (blue) vs. Metropolis (orange). Thin lines: individual samples; thick lines: sample mean.

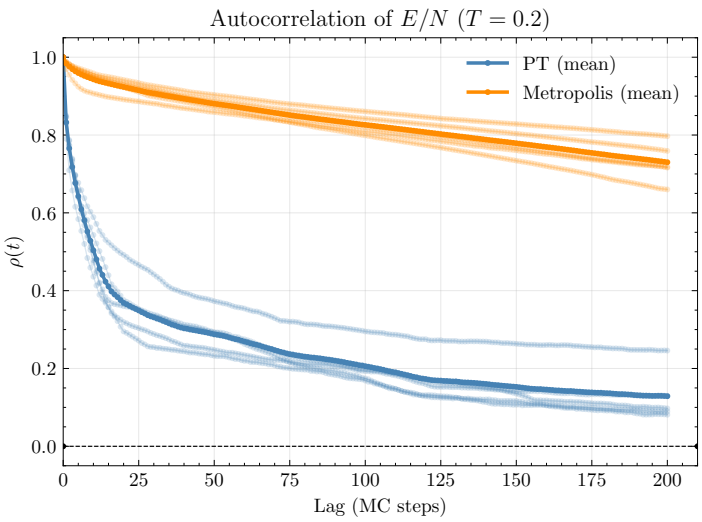

Figure 16: Normalised autocorrelation function $\rho(t)$ of E/N at $T = 0.2$: PT vs. Metropolis.

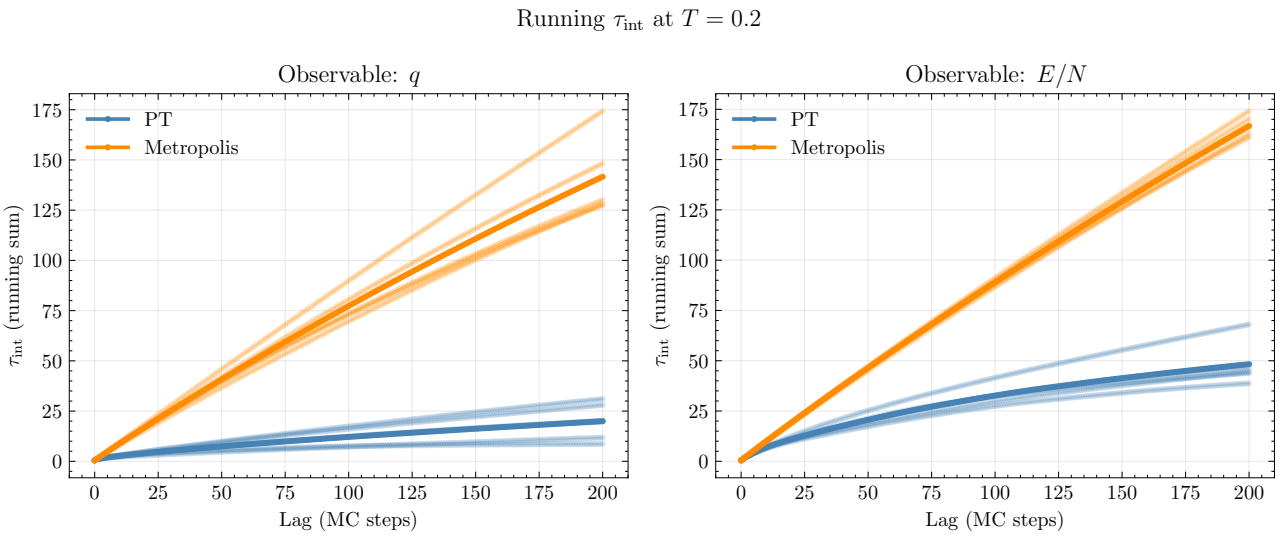

Figure 17: Running integrated autocorrelation time $\tau_{\text{int}}(t)$ vs. lag for q and E/N, PT vs. Metropolis, at $T = 0.2$.